JOURNAL OF GEOPHYSICAL RESEARCH, VOL. 101, NO. B5, PAGES 11,387–11,400, MAY 10, 1996

Paleomagnetism of upper Miocene ignimbrites at the Puna: An analysis of vertical-axis rotations in the Central Andes

Rubén Somoza and Silvia Singer
Laboratorio de Paleomagnetismo "Daniel Valencio", Departamento de Geología
Facultad de Ciencias Exactas y Naturales, Universidad de Buenos Aires, Buenos Aires, Argentina

Beatriz Coira
Instituto de Geología y Minería, Universidad Nacional de Jujuy, San Salvador de Jujuy, Argentina

Abstract. The origin of the rotations detected paleomagnetically in the Central Andes is controversial. Tectonic models proposed to explain it involve Late Cenozoic oroclinal bending (the Bolivian Orocline), and/or small-block rotations driven by oblique subduction. In this paper, we report paleomagnetic data from upper Miocene ignimbrites from the Puna, in the presumed southern limb of the orocline. These rocks showed high unblocking temperature and high coercivity magnetization. The resulting pole position (latitude 85.7°S, longitude 80.5°E, A_{95}=7.9°, K=22, N=17) indicates that the study area has not been significantly affected by vertical-axis rotation since late Miocene onward. This implies that the whole Puna has not undergone significant regional rigid-body rotation during the same time span. A farther analysis based on Neogene paleomagnetic data from 31 Andean localities distributed from 11°S to 31°S shows the presence of rotated and unrotated areas. Systematic counterclockwise rotations in Perú and northern Bolivia are observed, whereas clockwise rotations are present in southern Bolivia, northern Chile, and perhaps northwestern Argentina. No systematic rotations are found farther south. The overall time-spatial distributions of these Neogene paleomagnetic data suggest that if orogenic bending occurred, it must have taken place before the middle Miocene. Thus the oroclinal hypothesis can not explain the rotations observed in middle Miocene rocks or younger ones, which also show orocline-like declination anomalies. We suggest that small-block rotations driven by distributed shear may be a single process that can account for the whole rotations in the Central Andes. It is hypothesized that horizontal shear in the Andean crust could be controlled by ancient structures.

Introduction

The Central Andes are regarded as an example of mountain building related to subduction of an oceanic plate beneath a continental margin [*Dewey and Bird*, 1970; *James*, 1971; *Jordan et al.*, 1983]. Between 14° and 28°S, this orogen can be subdivided into three main physiographic regions. From west to east, they include the Andean forearc, the Altiplano-Puna plateau, and the Sierras Subandinas (or Subandes), the latter being the foreland fold-thrust belt (Figure 1a). These regions are separated by two almost parallel high mountain chains: the Cordillera Occidental, which matches the presently active volcanic arc, and the Cordillera Oriental.

Although the morphological expression of the orogen is Late Cenozoic in origin, the locus of the main deformational episodes have shown significant time-spatial variations. For instance, pervasive deformation occurred during middle-late Eocene in central Perú [*Mégard*, 1987], late Eocene-early Oligocene in northern Chile [*Scheuber and Reutter*, 1992], late Oligocene-early Miocene in the Bolivian Altiplano [*Sempere et al.*, 1990], late Miocene in the Puna [*Coira et al.*, 1982; *Jordan and Alonso*, 1987], and Pliocene-Quaternary in the Subandes [*Gubbels et al.*, 1993]. Nevertheless, an overall eastward shift of the deformation during the Cenozoic can be observed [e.g., *Mégard*, 1987; *Mpodozis and Ramos*, 1990; *Marret et al.*, 1994; *Kennan et al.*, 1995].

An outstanding characteristic of the Central Andes is the sudden swing in topographic and structural trend near 18-19°S (the Arica-Santa Cruz deflection, Figure 1a). *Carey* [1958] suggested that this feature is the result of a bending of an originally straight orogen. Later, *Isacks* [1988] suggested that the Andean crust was thickened mainly by differential along-strike tectonic stacking during the Late Cenozoic, with maximum shortening occurring close to the Arica-Santa Cruz bend. Isacks suggested that shortening has accommodated differential along-strike convergence between the Andean forearc and the Brazilian shield, leading to oroclinal bending of the former.

Testing these oroclinal hypotheses has been the aim of many paleomagnetic studies in the Central Andes. Their results have shown a widespread pattern of declination anomalies, the origin of which remains controversial.

Copyright 1996 by the American Geophysical Union.

Paper number 95JB03467.
0148-0227/96/95JB-03467$09.00

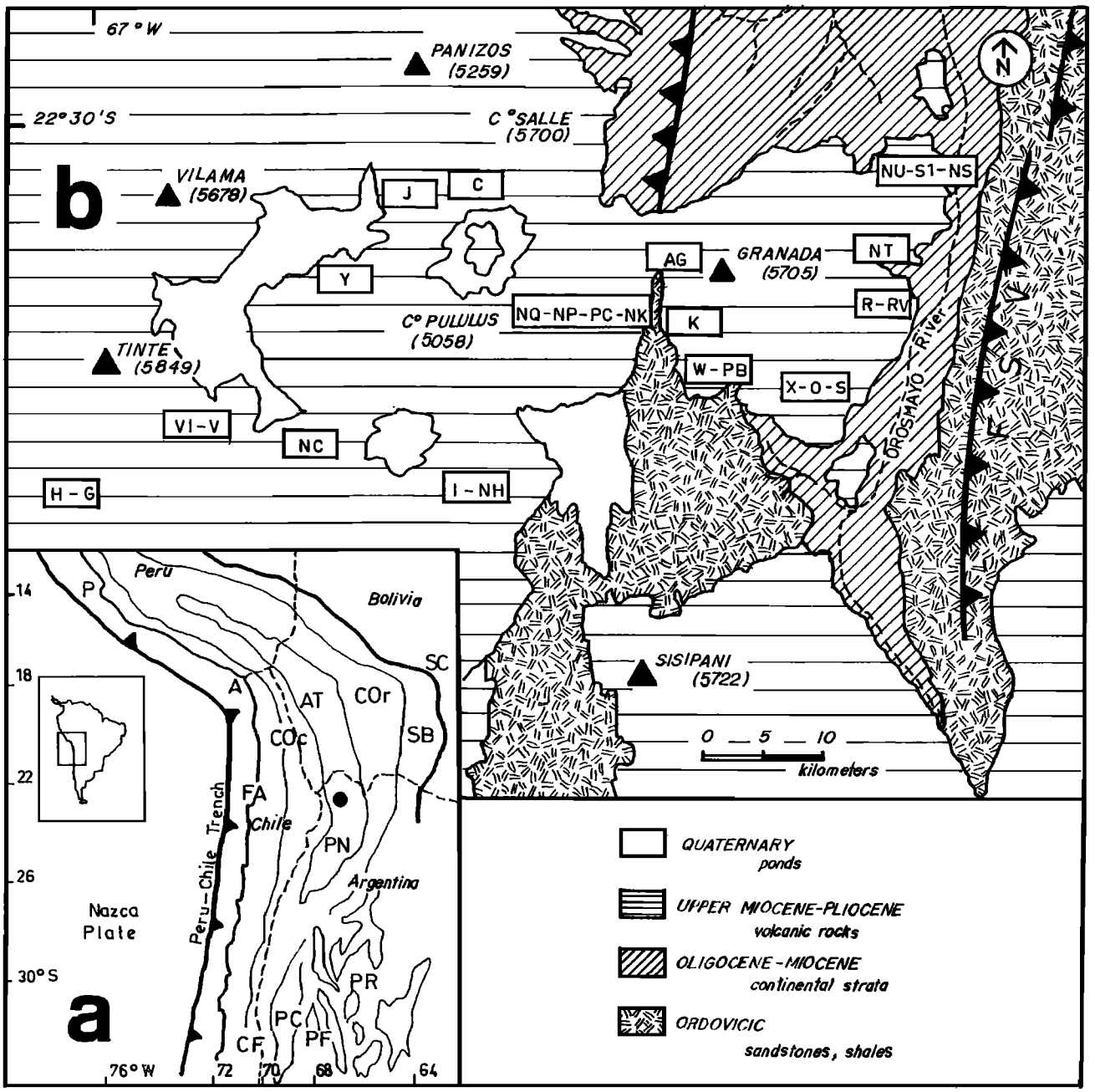


Figure 1. (a) Location of the mid-Andean chain showing its main morphotectonic provinces. AT, Altiplano; PN, Puna; COr, Cordillera Oriental; COc, Cordillera Occidental; FA, forearc; SB, Subandes (foreland fold and thrust belt); CF, Cordillera Frontal and Cordillera Principal; PC, Precordillera Argentina; PF, Pampean foreland; PR, Sierras Pampeanas. Solid circle indicates the study locality. (b) Geologic map of the sampling area [after *Turner*, 1982]. Sampling sites listed in Table 1 are shown. When more than one site is indicated in a window, the corresponding stratigraphic order is from left to right. Solid triangles represent main volcanoes (numbers indicate elevation above sea level). FSV is San Vicente Fault [*Herail et al.*, 1994]. Floor of the Orosmayo river valley is at 4000 m.

Counterclockwise rotations found by several studies in Perú were interpreted as evidence of block-like counterclockwise rotation of the Peruvian Andes [*Kono et al.*, 1985; *May and Butler*, 1985; *Macedo Sánchez et al.*, 1992a,b]. On the other hand, *MacFadden et al.* [1990, 1995] detected clockwise rotations in the southern Bolivian Andes and remarked that rotations are insignificant or absent in the areas close to the Arica-Santa Cruz bend. They concluded that the Central Andes can be divided into two nearly rigid blocks which have undergone opposite rotation about the Arica-Santa Cruz axis during the Late Cenozoic. However, some paleomagnetic studies [e.g. *Roperch and Carlier*, 1992; *Butler et al.*, 1995] have pointed out that a single and uniform mechanism of oroclinal bending does not explain the large declination anomalies observed in several Mesozoic and Lower Tertiary Andean rocks.

An alternative model to explain the observed declination anomalies comprises in situ small-block rotation resulting

from distributed shear related to oblique subduction [e.g., *Beck*, 1988; *Scanlan and Turner*, 1992; *Dewey and Lamb*, 1992]. In this model the Arica-Santa Cruz bend would be a primary feature of South America. In a recent contribution, *Beck et al.* [1994] combined in situ rotation of discrete blocks with a later oroclinal bending.

Late Cenozoic paleomagnetic data are crucial in understanding the kinematics of crustal deformation in the Central Andes, as it is during this time that oroclinal bending is thought to have occurred [e.g., *Isacks*, 1988; *Sempere et al.*, 1990 ; *MacFadden et al.*, 1995]. In this paper we report paleomagnetic results from upper Miocene-Pliocene ignimbrites of the Puna, which indicate no evidence of significant vertical axis rotation in the study area since late Miocene.

Furthermore, we have gathered the available paleomagnetic data of upper Tertiary rocks from widespread localities in the Central Andes. These data enabled us to obtain a picture constrained in time and space of the Neogene declination anomalies, which can be analyzed in the light of the two models (orocline versus small-block rotation).

Oroclinal bending involves rotation of a large block of crust about a fixed pivot point, which must then produce uniform and coeval declination anomalies throughout the entire region. In contrast, small-block rotation models predict that the amount (or also absence) of rotation in individual blocks to some extent are controlled by local geological conditions. In this last case, the occurrence of regional horizontal shear can control the sense of the rotations, which should be coeval with local crustal deformation.

The analysis suggests that if it is to be reconcilied with the considered paleomagnetic data, orogenic bending must be older than middle Miocene. However, we prefer a mechanism that involves local small-block rotation driven by distributed shearing, as it can account for the entire paleomagnetic data set. Vertical-axis rotations can be regarded as a component of the variable (in terms of space and time) deformation in the orogen. We hypothesize that horizontal shear in the Andean crust could be induced by convergence partitioning along pre-Andean structures.

Paleomagnetism of Upper Miocene Ignimbrites from the Puna

Geologic Setting

The sampling area is located in the northwestern Puna, on the east side of the present volcanic arc (Figure 1b). Slightly metamorphosed Ordovician shales and sandstones, tightly folded around N-NE trending axes, constitute the oldest rocks that crop out in the region [*Turner and Mendez*, 1979; *Cladohuos et al.*, 1994].

Widespread clastic strata were deposited on a broad alluvial plain during the Oligocene. This clastic material was supplied by erosional remnants of a late Eocene orogenic belt located west of the Puna and possibly also from uplifted pre-Tertiary rocks on the east [*Coira et al.*, 1982; *Jordan and Alonso*, 1987].

The beginning of the first magmatic phase in the Puna, at early-middle Miocene, coincides with the end of the Pehuenche deformational phase [*Coira et al.*, 1982, 1993], and its end with the end of the pervasive late Miocene Quechua deformational phase [*Coira et al.*, 1982].

After the late Miocene, the main deformational activity in this part of the Andes shifted eastward, to the Cordillera Oriental and Sierras Subandinas [*Isacks*, 1988; *Gubbels et al.*, 1993], at a time when a major volcano-tectonic province developed in the Altiplano-Puna [*Baker and Francis*, 1978; *de Silva*, 1989; *Coira et al.*, 1993]. This last volcanic episode marked the onset of strong ignimbritic eruptions in the study area. Post ignimbrite volcanic activity occurred during the Pliocene-Quaternary, building stratovolcanoes and dacitic domes.

The study ignimbrites are included in the upper Miocene-Pliocene Lipiyoc Formation [*Turner*, 1982]. They constitute a volcanic sequence that rest on Ordovician basement, slightly tilted Oligocene(?) strata and synorogenic Miocene strata. Radiometric ages of some of the studied rocks are around 8-9 Ma (see below). The Lipiyoc ignimbrites are flat lying, locally deformed by minor (<2 m displacement) strike-slip and normal faults [*Cladouhos et al.*, 1994]. Thus the study area was not affected by important post-Lipiyoc ignimbrites tectonic movements, other than regional ascension related to the uplift of the Puna [*Isacks*, 1988; *Gubbels et al.*, 1993].

Sampling, Treatment and Data Analysis Procedures

Paleomagnetic studies were carried out on 189 samples collected from 27 sites. Nine of these sites comprise the Vilama (five sites) and Atana (four sites) ignimbrites (see Table 1 and Figure 1b). The remainding 18 sites represent 17 cooling units, defined through petrological and geochemical studies [*Somoza et al.*, 1994], as well as field observations, such as cooling breaks and paleosurfaces between several units. Samples were collected using a portable gasoline-powered drill and were oriented in situ with both magnetic and solar compasses.

Eight ignimbrites were sampled in the Orosmayo river zone (Figure 1b), the lowest (NU site) being undated. At a higher stratigraphic level, a poorly welded ash flow (X site) yields a radiometric age of 8.9 ± 0.2 Ma (AQUATER, unpublished report, 1979). Some of these eastern ignimbrites are overlain by the unsampled 6 Ma Panizos Ignimbrite [*Ort*, 1993]. Ignimbrites sampled on the west side of the Cerro Granada (Figure 1b) constitute a thick sequence of about 350 m [*Coira*, 1990], which are overlain locally by either the 8.5 ± 0.2 Ma Vilama Ignimbrite or the 5 Ma lavas of the Cerro Granada platform [*Coira et al.*, 1993]. It is possible that the lower ignimbrites of this local sequence (sites W and AG) are the oldest units considered in this study. Farther west of the Cerro Granada, the samples of the Vilama Ignimbrite were obtained from widely separated sites (Figure 1b; sites PC, VI, NC, H, and I). This last ignimbrite is locally overlain by the Cerro Salle and Cerro Pululus ignimbrites (Figure 1b; sites C, NK, and Y). The youngest sampled unit, which is correlated with the 4 Ma Atana Ignimbrite [*Gardeweg and Ramirez*, 1987], shows isolated outcrops in the western study area (Figure 1b; sites J, NH, G, and V).

One to three standard specimens from each core sample were cut in the laboratory. Detailed alternating field (AF) and/or thermal demagnetization were performed using a Schonsted TSD-1 furnace and a rotating alternating field demagnetizer built at Buenos Aires University [*Vilas*, 1966]. Bulk magnetic susceptibility was measured after each heating

step in order to monitor possible magnetic mineral transformations. Remanence measurements were made on a Schonsted SSM2 and a modified Digico spinner magnetometers, both computer-interfaced. Anisotropy of magnetic susceptibility (AMS) measurements were made on a Kappabridge KLY-2 susceptibility bridge.

Magnetic directions were plotted on vector end point diagrams to identify discrete magnetic components, which were analyzed using principal component analysis [*Kirschvink*, 1980]. Directions were not accepted unless mean angular deviation (MAD) values were lesser than 10°. Most MAD values were less than 5°. Site and pole means were computed using standard *Fisher* [1953] statistics. Population analysis was done using the MAG88 program [*Oviedo*, 1989]. The Lipiyoc ignimbrites have undergone no discernible tectonic disturbance, and thus no structural corrections were applied.

Results

The natural remanent magnetization (NRM) intensity ranged from 0.005 A/m (site S1) to 18 A/m (some samples from site W), although it was generally between 0.1 and 1.2 A/m. The demagnetization behavior of the samples from most sites was straightforward (Figure 2), a single component of magnetization with high unblocking temperature and high coercivity was isolated, sometimes after removal of a soft component at no more than 300°C or 20 mT. Samples from site S1 showed unstable magnetization (e.g., Figure 3, specimen S002F1), so this site was excluded from farther study.

Samples from site Y and several samples from the Vilama Ignimbrite showed multicomponent magnetization. AF demagnetization on specimens from these samples failed to determine a final stable vector, but thermal demagnetization permitted the isolation of a high temperature component (e.g., Figure 3, specimens VI010A1, Y001T1).

Several samples from site W showed two components of magnetization with overlapping of both blocking temperature and coercivity spectra, whereas other samples from the same site showed univectorial behavior. In this case, demagnetization circles and stable final directions were combined to determine the site-mean direction [*McFadden and McElhinny*, 1988].

Isothermal remanent magnetization (IRM) experiments, performed on the most representative samples from these ignimbrites, usually showed efficient acquisition of magnetization and saturation at 250-400 mT. The IRM curves (Figure 4) suggest that the predominant carrier of the remanence is (Ti poor?) magnetite. Unblocking temperatures slightly higher than 580°C suggested the additional presence of some (Ti?) hematite in some sites. On the other hand, no complete saturation at 1500 mT indicated that hematite was the main carrier at C, Y, and Vilama Ignimbrite sites.

Bulk volume susceptibility ranged from 200 to 5000 10^{-6} SI, whereas the degree of anisotropy ranged from 0.9 to 2.1%. AMS measurements (unpublished data) showed that the magnetic anisotropy is low, with a lineation of around 0.5% and a magnetic foliation typically of 2%. Most sites showed minimum susceptibility axes (K_3) tightly grouped in a subvertical orientation, with more than 95% of all sites having K_3 axes within 20° of vertical.

Some selected thermally demagnetized specimens (which showed their magnetic foliation plane at near 45° from their Z axis) were heated at 600°C in the presence of a magnetic field of 5 mT, parallel to the Z axis of the specimens. The acquired magnetization in such cases was not deflected toward the foliation plane but was parallel to the applied magnetic field.

Interpretation of the Results and Their Tectonic Significance

Early magnetization in ignimbrites can be partially or totally altered after deposition by several intrinsic processes such as devitrification or plastic deformation after cooling through the blocking temperature [e.g., *Geissman*, 1980; *Rosembaum*, 1986].

The weak linear fabric shown by the AMS results argues against the possibility that the Lipiyoc ignimbrites have undergone important postemplacement flowing [*Wolff and Wright*, 1981].

Petrographic studies suggested that most of these ignimbrites had high emplacement temperatures. Most samples with a single component of magnetization showed inclinations close to the expected range for the present dipole field at Puna latitudes (Figure 5), suggesting that processes such as welding and/or compaction occurred entirely prior to the acquisition of the magnetization. Sites X and R showed a very shallow southward magnetization (Figure 2, specimen R001B2, and Figure 5). However, the relatively small lithostatic load (thickness of few tens of meters) and the nonwelded nature of these rocks suggest that compaction-related grains rotation are also unlikely to have occurred here.

Samples from Cerro Pululus (site Y, Figure 1b) showed a reverse (not well determined) low-temperature component and a normal high-temperature component. This site underwent later fumarolic processes, probably in the Pliocene, with development of a casiterite-hematite hydrothermal assemblage [*Coira*, 1990]. We interpret the high-temperature component at this site as a chemical magnetization related to low-temperature hydrothermal alteration.

The origin of the scattered softer component detected in some samples from Vilama Ignimbrite is not well understood. The high-temperature component determined for each site of this unit is interpreted as a primary magnetization, possibly originated by oxidation during cooling.

Thus we interpret the characteristic magnetizations of these ignimbrites as genuine records of the Late Cenozoic geomagnetic field at the Puna. The site mean directions are summarized in Table 1 and illustrated in Figure 5. To avoid oversampling of the unusually large paleosecular variation (PSV) observed in the Vilama and Atana units (figure 5), the corresponding sites for each of these were averaged to yield two single virtual geomagnetic poles (VGP). The same procedure was applied to the R and X sites (see Table 1). A mean pole position (LMPN, latitude 85.7°S, longitude 80.5°E, A_{95}=7.9°, K=22) was calculated by averaging 17 VGPs (Table 1).

The angular dispersion of VGPs (17.2°) is close to the 18° model value for dispersion due to PSV calculated for this latitude for the interval 5-22.5 Ma [*McFadden et al.*,

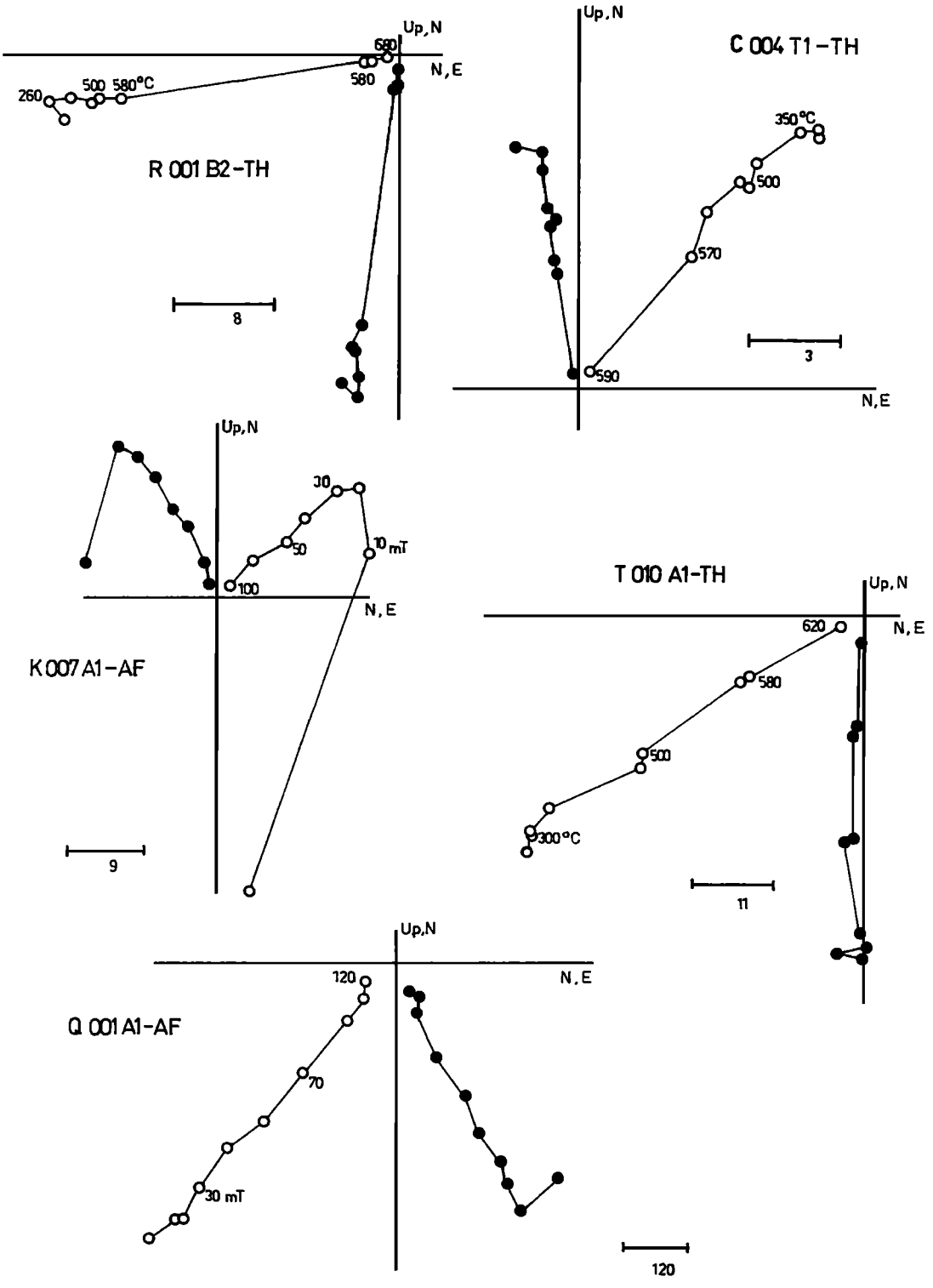


Figure 2. Vector component diagrams of progressive demagnetization for samples which show a single component of magnetization. Open (solid) circles indicate projection onto the vertical (horizontal) plane. Results of alternating field (thermal) demagnetization are indicated by "AF" ("TH") following the label of the specimen. Numbers adjacent to data points indicate peak AF (TH) demagnetization in milliteslas (degrees Celsius). Bars indicating the scale for magnetization intensity (in milliamperes) are also shown.

1991]. This together with the presence of cooling breaks and subaereal exposition surfaces between several units suggest that PSV has been adequately averaged. Nevertheless, the possibility of inadequate PSV's sampling in volcanic sequences is a problem that should always be taken into account [e.g., *Mankinen*, 1989].

The LMPN pole is indistinguishable from the present geographic axis, usually assumed as coincident with the Late

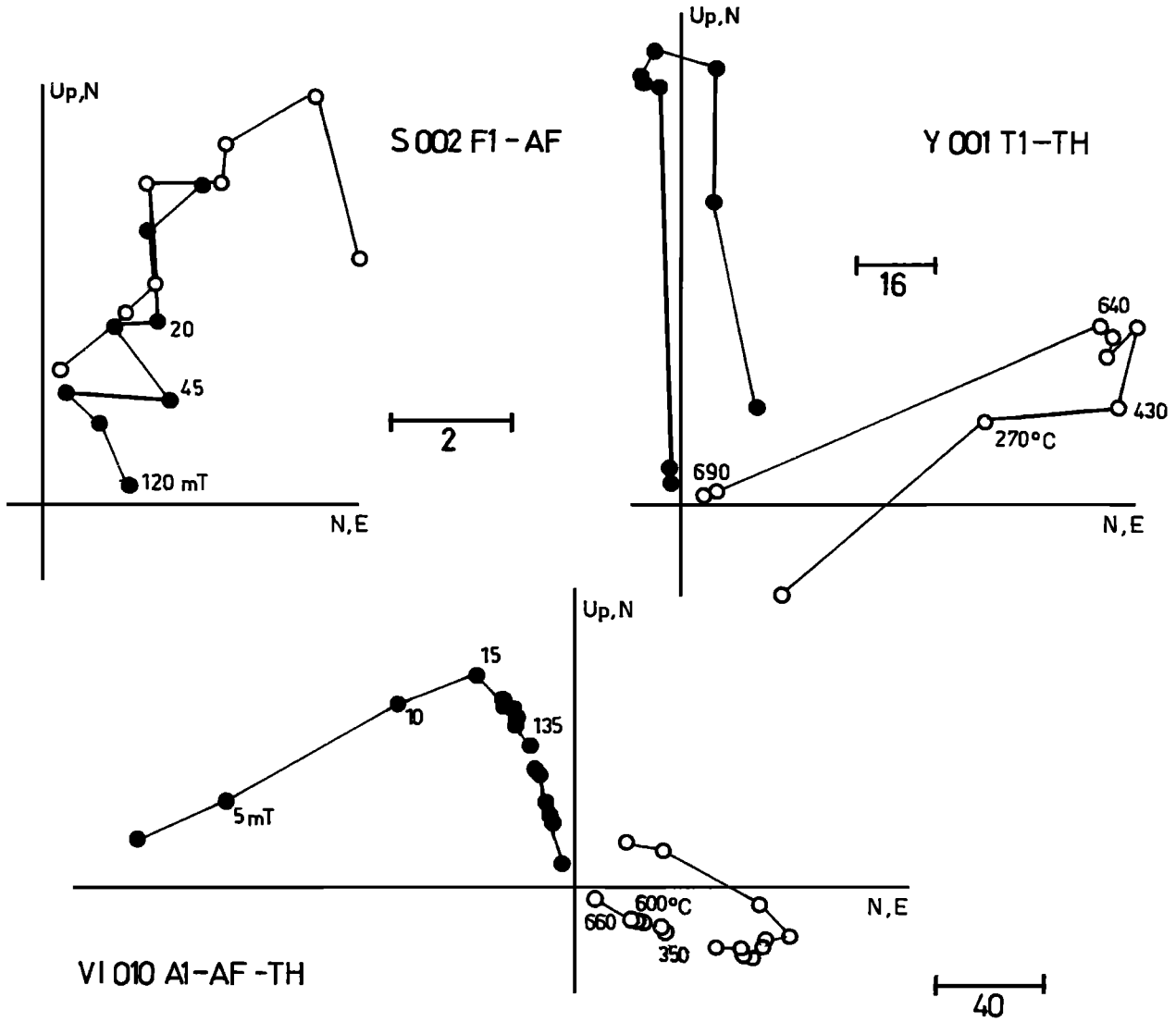


Figure 3. Vector component diagrams of progressive demagnetization for samples which show poorly defined demagnetization trajectory (S002F1) and multicomponent magnetization (Y001T1, VI010A1). Conventions are the same as for Figure 2.

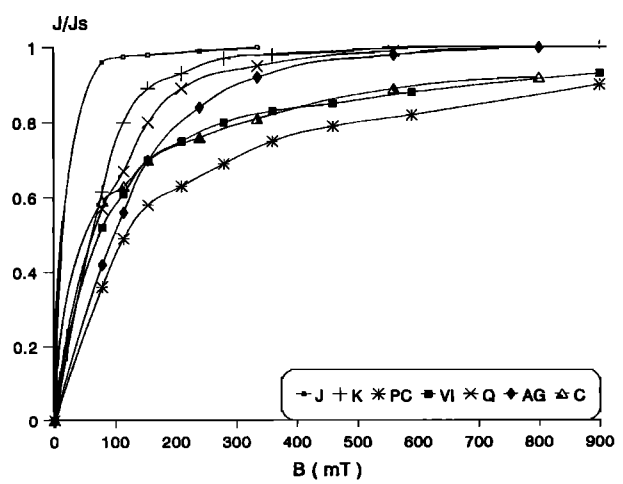


Figure 4. Isothermal remanent magnetization (IRM) acquisition curves for representative samples from the Lipiyoc ignimbrites. J/Js is normalized IRM; B is direct inductance (in milliteslas). Letters in the label of samples identify the corresponding site (see Table 1).

Cenozoic reference pole for South America [e.g., *MacFadden et al.*, 1995]. This indicates no evidence of significant vertical axis rotation in the study area since late Miocene.

On the other hand, considering that nearly 100 km of crustal shortening have been accommodated in the Subandean fold-thrust belt (eastward of the northern Puna) during the past 10 m.y. [e.g., *Gubbels et al.*, 1993], a direct implication of the paleomagnetic data here presented is that the youngermost phase of shortening in the Central Andes did not cause significant regional, rigid-body rotation of the whole Puna.

Tectonic Analysis of the Late Tertiary Paleomagnetic Data for the Central Andes

Data Compilation and Reference Pole Determination

It usually is assumed that if oroclinal bending has occurred in the Andes, at least part of it (if not all) must have taken place during Late Cenozoic time [e.g., *Isacks*,

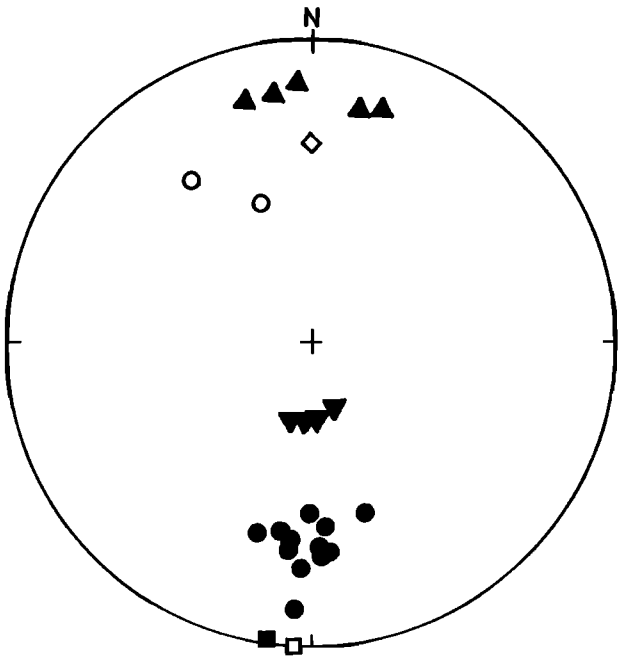


Figure 5. Equal-area projection of site mean magnetization, with solid (open) symbols indicating lower (upper) hemisphere. Squares, triangles, and inverted triangles indicate X-R, Vilama, and Atana ignimbrites sites, respectively (see Table 1). Site Y (Table 1) was neither considered in pole calculations nor represented in this Figure. Diamond shows the direction of the present dipole field at the study area.

1988; *Macedo Sánchez et al.*, 1992b; *Beck et al.*, 1994; *MacFadden et al.*, 1995]. To test this hypothesis, we gathered the available Late Tertiary Andean paleomagnetic data from several studies (Table 2). All data that we used have been obtained as a result of adequate demagnetization procedures and were interpreted as reliable records of the ancient geomagnetic field by the different sources. It must be pointed out that PSV may not be adequately averaged in some of the poles listed in Table 2, such as those based on only a few independent site-mean directions. As a first approach, we neglected possible problems derived from PSV in order to obtain a more complete time-spatial paleomagnetic coverage in the orogen. A more thorough tectonic analysis would be possible when greater paleomagnetic information about the Central Andes is available.

Most paleopoles from the Pampean foreland and the southern Puna were used after *Marret* [1990]. *Macedo Sánchez* [1993] obtained paleomagnetic data belonging to widely distributed sites located in the Peruvian forearc, Cordillera Occidental, and Altiplano. From this study we selected those areas where three or more paleomagnetic sites of Upper Tertiary rocks were available, enabling us to compute a mean pole position for each one. Thus five additional Peruvian areas were represented by paleomagnetic data: Acos, Chosica-Matucana, Villatambo, Posco, and Moquegua (Table 2).

A particular problem in tectonic paleomagnetic analysis for the Central Andes is the poorly determined South American apparent polar wander path. This is worsened by the commonly moderate to low amount of calculated Andean rotations.

Two paleomagnetic studies have been performed on South American Upper Tertiary rocks away from the Andean chain. *Creer and Valencio* [1969] reported paleomagnetic results from Upper Tertiary basalts outcropping in the southern Andean foothills, Argentina. *Richardson and Watkins* [1967] reported results from the 9 Ma volcanic and subvolcanic rocks at Fernando Noronha island, in the Brazilian equatorial Atlantic. We selected from these studies all site means with α_{95} lesser than 15°, obtaining seven normally and six reversely magnetized sites. A mean late Miocene pole position was calculated by averaging VGPs from these sites (LMSA, latitude 87.4°S, longitude 339.2°E, $A_{95}=5.1°$, $K=66$, $N=13$), which is indistinguishable from the present geographic pole. It is remarkable that the mean of the six north VGPs is separated from the inverted mean of the seven south VGPs by only 1.4° arc.

Main Characteristic of the Paleomagnetic Rotations

The LMSA paleopole was used as Late Cenozoic South American plate reference pole. The calculated rotations for Upper Tertiary rocks from the Central Andes are listed in Table 2. Figure 6a illustrates their classification and geographic distribution. These results differ slightly from other studies in which the present geographic pole was used as reference.

Examination of Figure 6a suggests that (1) not all Neogene paleomagnetic vectors are significantly rotated; (2) when rotations are significant, they are counterclockwise in the Peruvian and northern Bolivian Andes, and clockwise farther south, up to 23°S; (3) between 23°S and 27°S the Puna's region is not represented by paleomagnetic data, although preliminary results from Oligocene rocks at 26°S [*Singer et al.*, 1994] suggest the presence of both clockwise rotated and unrotated areas; (4) at 27°S (southernmost Puna), rocks at Corral Quemado (Figure 6a, datum 21) show significant clockwise rotation; and (5) south of 27°S (i.e., in the Pampean foreland), no systematic rotation pattern exists, as was previously noted by *Marret* [1990].

Previous studies have shown that paleomagnetic data from rocks older than Upper Tertiary in Perú, northern Bolivia, and northernmost Chile show a consistent pattern of counterclockwise rotations [e.g., *Roperch and Carlier*, 1992; *Macedo Sánchez et al.*, 1992a; *Roperch et al.*, 1993; *Beck et al.*, 1994], whereas Mesozoic rocks in northern Chile up to 28°-30°S show clockwise rotations. Mainly unrotated areas are found in the forearc farther south [*Somoza*, 1994]. Hence the spatial distribution of rotations from Neogene rocks (Figure 6a) resembles the rotation pattern defined by older rocks. This indicates the existence of systematic counterclockwise rotations in Perú, northern Bolivia, and northernmost Chile, and clockwise rotations in southern Bolivia, northern Chile, and perhaps also the Argentine Puna. Such distribution is referred here as the Central Andes rotation pattern (CARP).

To perform the temporal analysis of the rotations, we considered that the timing of rotation for an area must be younger than the youngermost radiometric age reported for rotated rocks in that area. Stratigraphic units without available radiometric data (see Table 2) were assigned an absolute age coincident with the mean value of the time span of the corresponding stage. This was done using the *Cande and Kent* [1992] time scale.

Table 1. Paleomagnetic Results

Site	N/No	*D*	*I*	α_{95}	*k*	Latitude	Longitude
			Orosmayo River Zone				
NU	5/6	179.0	43.4	4.8	250	87.0	311.3
NT	5/6	186.5	34.4	6.8	183	82.9	174.0
NS	4/4	177.0	34.2	9.0	88	85.3	76.0
S1	0/6	--	--	--	--	--	--
RV	4/4	195.5	35.1	8.9	109	75.2	194.1
X	3/4	182.0	-1.3	10.3	133	--	--
R	11/13	188.2	0.8	4.5	155	--	--
X-R[a]	14/17	186.8	0.3	4.5	78	66.7	130.9
O	5/5	182.8	13.3	8.5	83	74.0	123.6
S	4/5	186.0	31.6	4.0	953	82.2	160.8
			Volcán Granada Platform Zone				
NQ	5/6	161.6	41.2	4.9	247	73.0	16.0
AG	6/7	187.8	37.2	8.3	65	82.5	191.6
PB	9/9	182.2	27.5	4.4	135	81.8	128.6
K	8/10	177.3	31.0	6.6	72	83.7	89.3
W[b]	8/10	174.5	31.9	4.6	202	82.7	67.8
NP	5/5	174.7	39.7	10.9	50	85.1	22.0
			Cerro Salle-Pululus Zone				
C	5/5	340.0	-50.0	7.2	115	70.4	354.4
NK	3/3	324.4	-35.1	13.2	88	56.7	22.1
Y[c]	2/2	356.3	-21.3	--	--	--	--
			Vilama Ignimbrite				
PC	7/9	351.3	20.5	5.6	96	--	--
VI	6/7	016.4	21.8	8.1	66	--	--
NC	11/14	343.7	20.3	8.0	35	--	--
H	6/6	011.5	24.5	6.2	166	--	--
I	4/5	356.2	17.0	6.9	165	--	--
Mean	N=5	359.7	21.3	12.5	38	56.5	112.8
			Atana Ignimbrite				
V	17/20	175.4	68.8	4.6	60	--	---
G	9/9	158.7	72.0	2.7	353	--	--
NH	4/4	192.0	67.6	8.9	108	--	--
J	5/5	177.2	69.4	6.1	157	--	--
Mean	N=4	176.7	69.8	5.7	297	58.8	297.0

N/No are number of samples used in statistic/number of demagnetized samples; *D* and *I* are declination and inclination of site-mean magnetization; α_{95} is the 95% confidence level of site-mean direction; *k* is the dispersion parameter [Fisher,1953]; latitude and longitude are VGP coordinates (southern hemisphere); pole is 85.7°S, 80.5°E, A_{95}=7.9°, K=22, and N=17.

[a] Samples from the preceding two sites were averaged together (see the text)

[b] Site mean resulting from combination of demagnetization circles (5) and stable end point (3).

[c] Not used in mean pole calculation.

Figure 7a indicates that the timing of the rotations from Upper Tertiary rocks in the northern Central Andes is not uniform. On the other hand, rotations from Neogene rocks in the southern Central Andes show, irrespective of rock age, a roughly homogeneous clockwise signature (Figure 7b), which is broken by two concordant paleomagnetic data belonging to upper Miocene rocks.

Oroclinal Bending or Local Small-Block Rotations?

The CARP permits the orogen to be divided into two broad, continental-scale rotational domains, where paleomagnetic rotations were found in different rock types of different ages. We do not believe that causes other than tectonic ones developed such an arrangement. In the discussion that follows we test for the possible existence of a single process to account for the CARP.

Rotations in Neogene rocks from the Peruvian block do not show the synchronism required by a rigid-body, orogen-scale counterclockwise rotation, at least since mid-early Miocene onward (Figure 7a). No evidence of late Neogene rotation for Perú was already suggested by *Macedo Sanchez et al.* [1992b]. On the other hand, available paleomagnetic data constrain any clockwise rigid-body rotation of the southern Central Andes to have occurred during late Miocene, between deposition of the rotated 12 Ma Quebrada Honda section and the unrotated 9 Ma Lipiyoc ignimbrites (Figure 7b).

Different timing for rotation between north and south Central Andes requires an orogenic cross-strike transfer zone, with dextral displacement during early-middle Miocene to accommodate the northern block counterclockwise rotation and a later sinistral motion during late Miocene to accommodate the southern block clockwise rotation. Such

Table 2. Paleomagnetically Determined Rotations from Upper Terciary Rocks, Central Andes

Area	Location		Age,[a]	Pole					$R \pm \Delta R$
	Latitude °S	Longitude °E	Ma	Latitude	Longitude	A_{95}	N	Source	
1, Acos	11.2	283.3	19	85.2	211.4	9.1	3	1	06.9 ± 8.5
2, Chosica	12.1	283.5	21	65.5	005.6	12.8	5	1	-22.9 ± 11.4
3, Ocros	13.4	286.0	5	75.7	212.3	5.2	30	2	-12.9 ± 6.1
4, Pisco	14.5	284.5	6	87.2	096.0	5.7	10	3	01.4 ± 7.9
5, Villatambo	14.7	285.4	19	73.5	018.5	5.0	5	1	-14.7 ± 5.9
6, Posco	16.0	286.7	11	89.3	210.8	7.1	3	1	02.9 ± 7.3
7, Puno	16.0	290.0	6	80.0	038.5	15.4	6	3	-07.6 ± 13.4
8, Moquegua	17.0	289.2	LM	86.6	207.1	5.6	4	1	05.6 ± 6.4
9, Chama	17.0	291.7	M	77.5	012.2	11.5	9	4	-11.0 ± 10.2
10, Salla	17.2	292.3	26	82.7	350.5	7.5	58	5	-04.6 ± 7.7
11, Puesto Japones	17.5	291.7	M	71.3	040.1	4.0	3	4	-16.0 ± 5.4
12, Micaña	17.5	292.5	7	84.6	058.6	5.3	25	5	-02.5 ± 6.1
13, Inchasi	19.7	294.7	4	84.1	061.1	4.9	47	6	-03.0 ± 6.0
14, Quehua	20.0	293.0	8	75.6	211.0	8.5	25	7	17.3 ± 8.5
15, Lipez	21.7	293.4	UM	85.8	210.8	8.2	5	4	06.5 ± 8.4
16, Guadalupe	21.8	293.4	20	53.8	210.2	10.0	15	4	41.0 ± 9.7
17, Cerdas	21.8	293.3	15	80.6	207.2	6.5	24	7	12.2 ± 7.1
18, Quebrada Honda	22.0	294.5	12	73.5	212.3	4.0	79	5	19.8 ± 5.7
19, Lipiyoc	22.5	293.0	9	85.7	080.5	7.9	17	8	-0.4 ± 8.0
20, Paciencia	23.0	291.0	24	70.6	177.0	12.5	9	9	20.3 ± 11.2
21, Corral Quemado	27.2	293.1	4	73.4	202.9	3.7	--	10	20.6 ± 5.7
22, Río Guanchin[b]	27.7	292.8	4	83.1	266.0	7.0	--	10	05.8 ± 8.2
23, Quebrada Troya[b]	27.9	292.3	5	76.5	030.3	7.0	--	10	-12.6 ± 7.8
24, Las Juntas[b]	29.4	291.4	10	76.7	022.6	4.6	--	10	-12.9 ± 6.3
25, Río Mañero[b]	29.7	292.2	6	84.7	090.3	4.2	--	10	-0.5 ± 6.0
26, Angualasto[b]	30.0	290.5	10	82.9	023.2	9.7	9	11	-05.8 ± 8.0
27, Huaco Norte[b]	30.1	291.6	9	79.1	291.6	7.2	--	10	13.7 ± 8.0
28, Huaco[b]	30.1	291.6	3	81.8	180.1	6.0	92	12	11.4 ± 7.2
29, Río Jachal[b]	30.1	291.6	3	80.4	178.2	3.3	--	13	12.0 ± 5.6
30, Huaco Sur[b]	30.3	291.5	9	83.8	116.3	6.7	--	10	02.8 ± 7.6
31, Río Azul[b]	30.7	291.2	9	83.4	127.1	5.0	89	14	04.3 ± 6.5

Location indicates the geographic coordinates of the respective sampling area. Pole parameters (latitude, longitude, A_{95}, N) indicate latitude (°S), longitude (°E), confidence interval (in degrees), and number of sites. N are not available for data compiled by Marret[1990]. Sources of paleomagnetic data are 1, *Macedo Sanchez* [1993]; 2, *Heki et al.*[1985]; 3, *Tsunakawa et al.*[1987]; 4, *Roperch et al.*[1993]; 5, *MacFadden et al.*[1990]; 6, *MacFadden et al.*[1993]; 7, *MacFadden et al.*[1995]; 8, this study; 9, *Hartley et al.*[1992]; 10, after *Marret* [1990]; 11, *Ré and Barredo*[1993];12, *Johnson et al.*[1984]; 13, *Johnsson* [1984]; and 14, *Jordan et al.*[1990]. R and ΔR are rotation (clokwise positive) and confidence interval [after *Beck et al.*,1986] with respect to the LMSA pole (see text).

[a] When possible minimum probable age for significantly rotated rocks and maximum for unrotated ones. M, LM and UM indicate Miocene, lower Miocene, and upper Miocene, respectively.

[b] Data from the Pampean foreland.

a transfer zone is not obvious in either the Altiplano or the forearc. This suggests that, if it occurred, orogenic bending must be older than middle Miocene. On the other hand, concordant paleomagnetic data from Peruvian coastal areas suggest that crustal shortening younger than late Miocene did not cause paleomagnetically determinable rotation of the Pisco-Arica segment (which, in turn, defines the greatest orogenic deflection, Figure 6a).

These observations suggest that the rotations of the CARP found in middle Miocene and younger rocks, although showing orocline-like declination anomalies, must be the result of local deformation. Thus the possibility exists that the rotations shown by rocks older than middle Miocene had also been produced by local deformation. However, two apparent problems appear to argue against this last possibility. These are (1) the widespread presence of rotations in Mesozoic and Tertiary Andean rocks, and (2) the similar magnitude of these rotations [e.g., *Macedo Sánchez et al.*, 1992a,b; *Beck et al.*, 1994]. We believe that these apparent constraints for small-block rotation models are not severe when regarded in the Andean tectonic frame.

In a compressional setting (like the Andes), deformation may stop when the crust has been thickened enough and then will shift elsewhere [*McKenzie and Jackson*, 1983; *Park*, 1988]. Small-block rotations in the Central Andes may have taken place during pulses of deformation, which occurred at different times in different areas (for the timing of deformation, see discussion by *Marret et al.* [1994]). Most areas west of the Cordillera Oriental have been affected by at least one important pulse of crustal deformation older than latest Miocene. This would increase the likelihood of the presence of local vertical-axis rotation in rocks older than upper

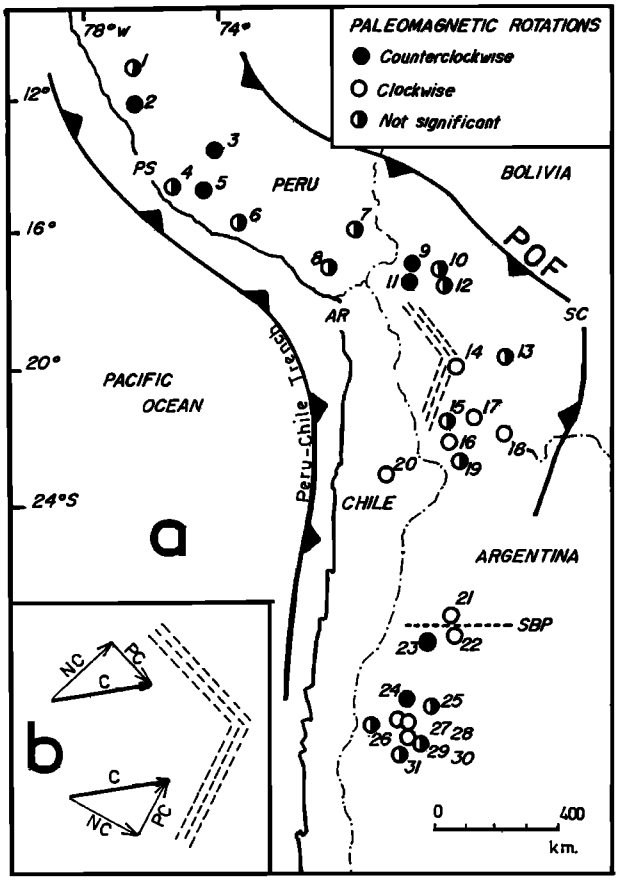


Figure 6. (a) Geographic distribution and classification (according to Table 2) of Late Tertiary paleomagnetic data for the Central Andes. PS, SC, and AR denote Pisco, Santa Cruz and Arica localities, respectively. POF and SBP indicate the present orogenic front and the southern boundary of the Puna, respectively. Numbers besides symbols identify the considered areas, as they are listed in Table 2. A domain of counterclockwise rotations can be defined in the northern Central Andes, whereas a domain of clockwise rotations can be defined in the southern Central Andes, up to the SBP. No systematic rotations are found farther south. The boundary between the counterclocwise and clockwise domains is arbitrarily placed to be coincident with the AR-SC line. Dashed lines show the abrupt change of the gravity lineaments and structural trend in the Bolivian Altiplano (see the text). (b) C represents the Late Cenozoic Nazca-South America convergence vector [after *Pardo Casas and Molnar*, 1987]. Dashed lines resemble the structural trend in the Altiplano, as shown in Figure 6a. The possible partition of C into components normal (NC) and parallel (PC) to such a structural trend is shown. Note that sinistral shear must be produced where the structural trend is NW, whereas dextral shear is expected where it is NE. This is in agreement with the observed rotations (Figure 6a).

Miocene in that region. However, this latter possibility should decrease in the youngermost rocks, because from late Miocene onward the main deformational activity is located in the Subandes. This is in agreement with the timing shown in Figure 7. Unfortunately, Andean paleomagnetic data are sparse and poorly constrained in time to test if rotations roughly follow the overall eastward shift of the deformation during the Cenozoic. On the other hand, time-accumulated local rotation could be possible only when more than one tectonic pulse (without change in the sense of horizontal shear) had affected a specific area (for instance, *Hartley et al.* [1992]).

Hence we suggest that small-block rotations driven by distributed shear provide a single process that could account for the CARP. The regional shearing can be related to convergence kinematics [e.g., *Beck*, 1988; *Dewey and Lamb*, 1992; *Scanlan and Turner*, 1992]. *Pardo Casas and Molnar* [1987] found that oblique convergence occurred between the South American and Nazca plates during the Cenozoic. *Beck* [1988] noted that the tangential component of this convergence, which reverses at the Arica elbow, can account for the observed sense of the rotations.

A particular problem in Central Andean tectonics is how and to what extent the plate convergence vector is converted into strain in the overriding plate. The absence of major strike-slip faults in the back arc region, together with the widespread distribution of the CARP, suggest that horizontal shearing in the Andean crust must have been widely distributed in map view. *Isacks* [1988] assumed that the late Cenozoic convergence between the forearc and the foreland accommodates horizontal shortening and rotational shearing in the back arc. *Dewey and Lamb* [1992] suggested significant sinistral shear in northern Bolivia and Perú to accommodate the relative plate convergence during the past 5 m.y. A slightly dextral shear in the Puna was found for the Quaternary [*Marret et al.*, 1994]. However, the rotational shear driven agent for the CARP remains uncertain.

Consideration of the Basement Structures as a Controlling Factor on the Rotations

Many studies of the Central Andes suggested the occurrence of an important paleotectonic control on the Cenozoic deformation [e.g., *Jordan et al.*, 1983; *Baby et al.*, 1989; *Gubbels et al.*, 1993; *Allmendinger* et al., 1993]. In particular, *Somoza* [1994] showed that the presence of rotations in the Chilean Andes correlates with inherent characteristics of the overriding plate. Thus a paleotectonic factor in the control of the Andean rotations could be a workable hypothesis. In this section we show some interesting correlations on this subject.

The boundary between the two rotational domains for the Central Andes must be located in the Bolivian Altiplano. In this region, the trend of gravity anomalies change from NW in the northern Altiplano to NE/N-NE farther south (Figure 6a), in agreement with the trend of Paleozoic and Cenozoic structures [*Cady and Wise*, 1992]. Isotopic composition of volcanic rocks indicates that different basement domains correspond to each one of those different structural trends [*Aitcheson et al.*, 1995]. Thus these latter may be a primary (pre-Andean orogeny) feature in the Altiplano.

Figure 6a shows that counterclockwise rotations are placed where this structural trend is NW, whereas clockwise rotations are present where it is NE. This suggests a possible mechanical relationship between small-block rotations in the Altiplano and the trend of ancient structures. The persistence of this trend of fractures in the northernmost Altiplano was noted by *Isacks* [1988] after inspection of computer digitized topographic maps. Farther south, in the Puna, it is also indicated by Ordovician structures [*Mon and Hongn*, 1988].

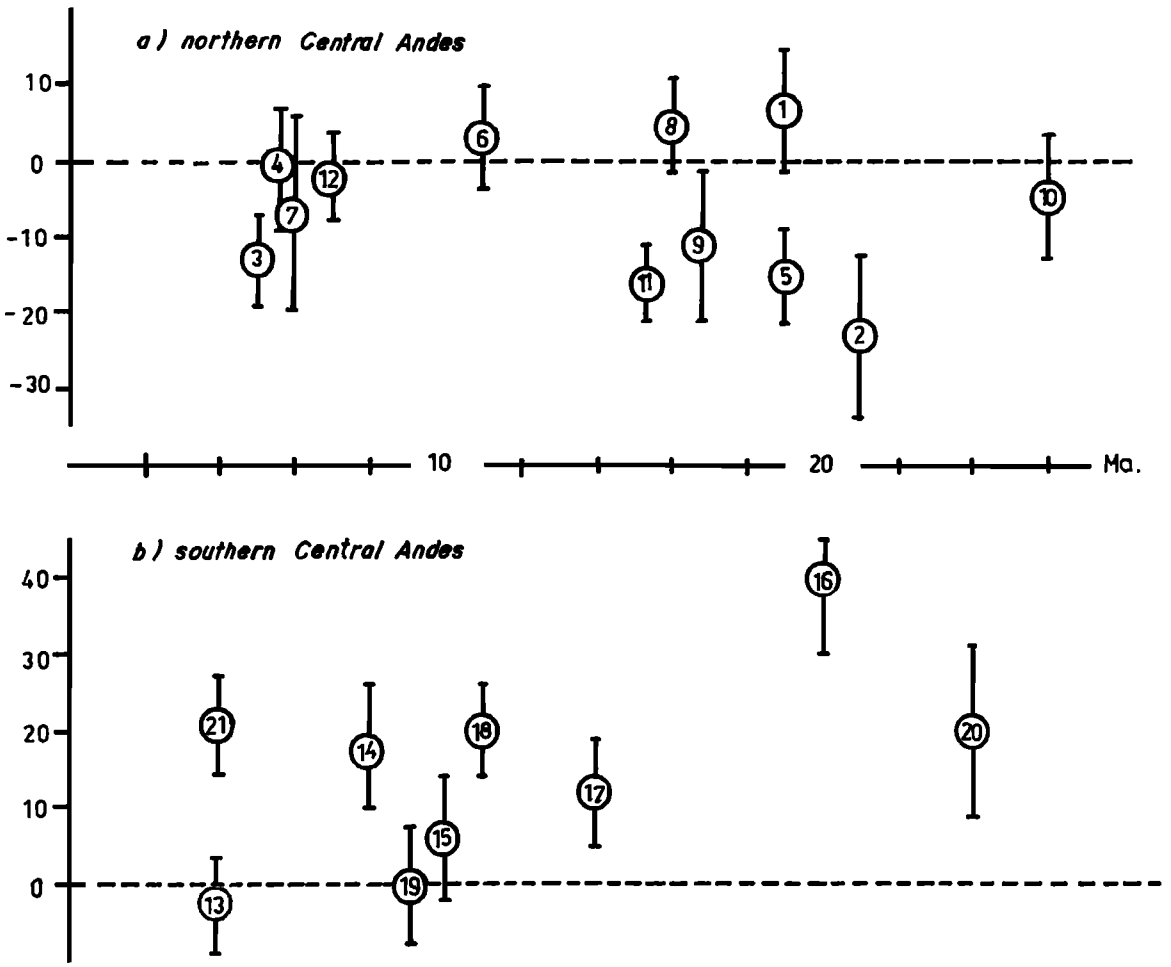


Figure 7. Ordinate is degrees of rotation (clockwise positive) found from Upper Tertiary rocks in the (a) northern and (b) southern Central Andes. Abscissa is rock age. Numbers into symbols indicate the respective area (see Table 2). Rotations, rock age, and 95% confidence intervals as in Table 2. Paleomagnetic data from the Pampean foreland were omitted in the plot. The boundary between the northern and southern Central Andes is assumed as in Figure 6a. Note that since late Miocene (about 10 Ma) onward the frequency of rotations appear to decrease, in agreement with the shift of the main deformation toward the Subandes (see the text). Datum number 21 is from rocks at the southern Puna, where crustal shortening remained active till the latest Pliocene-Pleistocene [*Allmendinger*, 1986; *Allmendinger et al.*, 1993].

The fact that convergence was oblique to ancient structural trends in the Central Andes implies the possibility that the strain resulting from the plate convergence may have been partitioned into components normal and parallel to such a structural trend (Figure 6b). This is because slip on preexisting crustal weaknesses may be more likely than formation of new fractures and also than continuous deformation. The sense and areal extension of the horizontal shear produced by such strain partitioning must be the ones expected to account for the observed sense of the rotations and their persistence far inland.

Similar strain partitioning has been suggested to explain the observed crustal deformation in several areas of the orogen. For instance, *Baby et al.* [1993] suggested that convergence partitioning in the Santa Cruz bend area (Bolivian Subandean zone) has been accommodated by normal shortening and parallel strike-slip with respect to the local structural trend, which in turn is controlled by Paleozoic paleogeography. *Marret et al.* [1994] noted that the main Late Cenozoic shortening phase in the Puna was not parallel to the plate convergence but normal to Ordovician structures. In this last case, the coeval presence of companion strike-slip motion is not obvious.

Concluding Remarks

In this paper we have presented paleomagnetic data from upper Miocene ignimbrites in the Puna, which seem to provide a genuine record of the late Miocene-Pliocene paleomagnetic field. Although these data may not adequately average paleosecular variation, the in situ group mean direction is indistinguishable from the expected Late Tertiary reference direction. This indicates no evidence of significant vertical axis rotation in the study area since late Miocene time, which implies that the northern Puna has not undergone significant regional rotation during that time span.

A tectonic analysis based on Neogene paleomagnetic data from 31 Andean localities showed the presence of rotated and unrotated paleomagnetic vectors. Those which show rotations, together with paleomagnetic data from Mesozoic

and Paleogene Andean rocks, allow one to distinguish two rotational domains: a counterclockwise domain defined in Perú, northern Bolivia, and northernmost Chile, and a clockwise one defined in southern Bolivia, northern Chile, and perhaps northwestern Argentina. Paleomagnetic vector rotations from both these two domains constitute the Central Andes rotation pattern (CARP).

We suggest that the dominant process to account for the CARP is small-block rotations driven by distributed shearing of the Central Andean crust. We show correlation between the CARP and ancient crustal structures in the Altiplano. It seems possible that this correlation can be more widely extended in the orogen. We hypothesize that the regional shearing to drive small-block rotations could be the result of convergence partitioning along preexisting crustal fractures.

Nevertheless, whether or not the orogen experienced a significant amount of oroclinal bending before middle Miocene, as well as if the arcuate shape of the Andean forearc is product of an oroclinal bending, still remains controversial. A widest test could be possible when more Andean paleomagnetic data become available. This should have important implications in the knowledge of the deformation experienced by this singular, subduction-related orogen; and in determining if the Arica elbow, as we know today, constitutes a primary or secondary feature of South America.

Acknowledgments. We thank V. Ramos, A. Rapalini, H. Vizán, M. Uliana, J. Cortés, and J. Vilas for helpful comments. E. Viviani, N. Durante, and S. Geuna prepared the illustrations. P. Iglesias LLanos, C. Siano and L. Rodríguez checked our English syntax, grammar, and spelling. A. Perez provided us his "Puna travel" experience. AMS measurements were performed in the Centre des Faibles Radioactivités, CNRS CEA (France) and in the Departament of Geological Sciences, University of California at Santa Barbara by means of cordial invitations from C. Laj and M. Fuller, respectively. Paleomagnetic data provided by O. Macedo Sánchez and B. MacFadden were useful in the development of this study. Careful reviews by Andrew Tomlinson, Myrl Beck, Pierrick Roperch, Bruce MacFadden, Bill MacDonald, Jimmy Diehl (Associate Editor), and an anonymous reviewer contributed to improvement of the manuscript. Our laboratory of paleomagnetism resides at the Universidad de Buenos Aires, which also provides us with the necessary elements. This research was supported by CONICET, PID-BID 320 and PID 504.

References

Aitcheson, S., R. Harmon, S. Moorbath, A. Schneider, P. Soler, E. Soria-Escalante, G. Steele, I. Swainbank, and G. Wörner, Pb isotopes define basement domains of the Altiplano, Central Andes, *Geology, 23*, 555-558, 1995.

Allmendinger, R., Tectonic development, southeastern border of the Puna Plateau, northwestern Argentine Andes, *Geol. Soc. Am. Bull, 97*, 1070-1082, 1986.

Allmendinger, R., T. Gubbels, B. Isacks, and T. Cladouhos, Lateral variations in Late Cenozoic deformation, Central Andes, 20-28°S (extended abstract), in *Symposium International Géodynamique Andine, 2nd.*, edited by ORSTOM, pp. 155-158, Oxford, England, 1993.

Baby, P., G. Herail, J. Lopez, O. Lopez, J. Oller, J. Pareja, T. Sempere, and D. Tufiño, Structure de la Zone Subandine de Bolivia: Influence de la géométrie des séries sédimentaires anté orogéniques sur la propagation des chevauchements, *C. R. Acad. Sci., 309*, 1717-1722, 1989.

Baby, P., B. Guillier, J. Oller, G. Herail, G. Montemurro, D. Zubieta, and M. Specht, Structural synthesis of the Bolivian Subandean Zone (extended abstract), in *Symposium International Géodynamique Andine, 2nd.*, edited by ORSTOM, pp, 159-162, Oxford, England, 1993.

Baker, M., and P. Francis, Upper Cenozoic volcanism in the Central Andes - Ages and volumes, *Earth Planet. Sci. Lett, 41*, 15-187, 1978.

Beck, M., Analysis of Late Jurassic-recent paleomagnetic data from active margins of South America, *J. South Am. Earth Sci., 1*, 39-52, 1988.

Beck, M., R. Burmester, D. Craig, C. Grommé, and R. Wells, Paleomagnetism of middle Tertiary volcanic rocks from the Western Cascade Series, *J. Geophys. Res., 91*, 8219-8230, 1986.

Beck, M., R. Burmester, R. Drake, and P. Riley, A tale of two continents: Some tectonic contrast between the Central Andes and the North American Cordillera, as illustrated by their paleomagnetic signatures, *Tectonics, 13*, 215-224, 1994.

Butler, R., D. Richards, T. Sempere, and L. Marshall, Paleoma gnetic determinations of vertical-axis tectonic rotations from Late Cretaceous and Paleocene strata of Bolivia, *Geology*, 23, 799-802, 1995.

Cady, J., and R. Wise, Gravity and magnetic studies, in *Geology and Mineral Resources of the Altiplano and Cordillera Occidental, Bolivia, U.S. Geol. Surv. Bull., 1975,* edited by D. Bleiwas and R. Christians, pp. 56-62, 1992.

Cande, S., and D. Kent, A new Geomagnetic Polarity Time Scale for the Late Cretaceous and Cenozoic, *J. Geophys. Res., 97*, 13,917-13,951, 1992.

Carey, S., The tectonic approach to continental drift, in *Continental Drift- A Symposium,* edited by S. Carey, pp. 178-355, Univ. of Tasmania Press, Hobart, Tasmania, 1958.

Cladohuos, T., R. Allmendinger, and B Coira, Late Cenozoic deformation in the Central Andes: Fault kinematics from the northern Puna, northwestern Argentina and southwestern Bolivia, *J. South Am. Earth Sci., 7*, 209-228, 1994.

Coira, B., Ignimbritas y lavas dacíticas del Cenozoico superior de la Puna jujeña y sus contenidos anómalos en Sn, Sb, Ag y Au, in *XI Congreso Geológico Argentino*, vol.1, pp. 299-302, San Juan, Argentina, 1990.

Coira, B., J. Davidson, C. Mpodozis, and V. Ramos, Tectonic and magmatic evolution of the Andes of northern Argentina and Chile, *Earth Sci. Rev., 18*, 303-332, 1982.

Coira, B., S. Mahlburg Kay, and J. Viramonte, Upper Cenozoic magmatic evolution of the Argentine Puna - A model for changing subduction geometry, *Int. Geol. Rev., 35*, 677- 720, 1993.

Creer, K., and D. Valencio, Palaeomagnetic and rock magnetic studies on the Cenozoic basalts from western Argentina, *Geophys. J. R. Astron. Soc., 19*, 113-146, 1969.

de Silva, S., Altiplano-Puna volcanic complex of the Andes, *Geology, 17*, 1102-1106, 1989.

Dewey, J., and J. Bird, Mountain belts and the New Global Tectonics, *J. Geophys. Res., 75*, 2625-2647, 1970.

Dewey, J., and S. Lamb, Active tectonics of the Andes, *Tectonophysics, 205*, 79-95, 1992.

Fisher, R., Dispersion on a sphere, *Proc. R. Soc. London, A, 217*, 295-305, 1953.

Gardeweg, M., and C. Ramírez, La Pacana caldera and the Atana Ignimbrite - A major ash-flow and resurgent caldera complex in the Andes of northern Chile, *Bull. Volcanol., 49*, 547-566, 1987.

Geissman, J., Paleomagnetism of ash-flow tuffs: Microanalytical recognition of TRM components, *J. Geophys. Res., 85*, 1487-1499, 1980.

Gubbels, T., B. Isacks, and E. Farrar, High-level surfaces, plateau uplift, and foreland development, Bolivian Central Andes, *Geology, 21*, 695-698, 1993.

Hartley, A., E. Jolley, and P. Turner, Paleomagnetic evidence for rotation in the Precordillera of northern Chile: Structural constraints and implications for the evolution of the Andean forearc, *Tectonophysics, 205*, 49-64, 1992.

Heki, K., Y. Hamano, M. Kono, and T. Ui, Paleomagnetism of Neogene Ocros dyke swarm, the Peruvian Andes: Implication for the Bolivian orocline, *Geophys. J. R. Astron. Soc., 80*, 527-534, 1985.

Herail, G., P. Baby, and P. Soler, El contacto Cordillera Oriental-Altiplano en Bolivia: Evolución tectónica, sedimentaria y

geomorfológica durante el Mioceno, in *Actas 7° Congreso Geológico de Chile*, vol.1, pp.62-66, Concepción, Chile, 1994.

Isacks, B., Uplift of the Central Andean plateau and bending of the Bolivian Orocline, *J. Geophys. Res., 93*, 3211-3231, 1988.

James, D., Plate tectonic model for the evolution of the Central Andes, *Geol. Soc. Am. Bull., 82*, 3325-3346, 1971.

Johnson N., T. Jordan, P. Johnsson, and C. Naeser, Magnetic polarity stratigraphy, age and tectonic setting of fluvial sediments in an eastern Andean foreland basin, San Juan province, Argentina, *Spec. Publ. Int. Ass. Sedimentol., 8*, 63-75, 1986.

Johnsson, P., Magnetic polariry stratigraphy and age of the Río Jachal and Mogna Formations, San Juan province, Argentina, M.A. thesis, Dartmouth College, Hannover, Ind., 1984.

Jordan, T., and R. Alonso, Cenozoic stratigraphy and basin tectonics of the Andes mountains, 20°-28° south latitude, *AAPG Bull., 71*, 49-64, 1987.

Jordan, T., B. Isacks, R. Allmendinger, J. Brewer, V. Ramos, and C. Ando, Andean tectonics related to geometry of subducted Nazca plate, *Geol. Soc. Am. Bull., 94*, 341-361, 1983.

Jordan, T., P. Rutty, L. McRae, J. Beer, K. Tabbutt, and J. Damanti, Magnetic polarity stratigraphy of the Miocene Río Azul section, Precordillera thrust belt, San Juan province, Argentina, *J. Geol., 98*, 519-539, 1990.

Kennan, L., S. Lamb, and C. Rundle, K-Ar dates from the Altiplano and Cordillera Oriental of Bolivia: Implications for Cenozoic stratigraphy and tectonics, *J. South Am. Earth Sci., 8*, 163-186, 1995.

Kirschvink, J., The least-squares line and plane and the analysis of paleomagnetic data, *Geophys. J. R. Astron. Soc.*, 62, 699-718, 1980.

Kono, M., K. Heki, and Y. Hamano, Paleomagnetic sudy of the Central Andes: Counterclockwise rotation of the Peruvian block, *J. Geodyn., 2*, 193-209, 1985.

Macedo Sanchez, O., Etude paléomagnétique des formations Crétacées et tertiaires des Andes Centrales du Peru: Role des rotations dans la formation des déflexions andines, Ph.D. thesis, Univ. Paris XI, 1993.

Macedo Sanchez, O., J. Surmont, C. Kissel, P. Mitouard and C. Laj, Late Cenozoic rotation of the Peruvian Western Cordillera and the uplift of the Central Andes, *Tectonophysics, 205*, 65-77, 1992a.

Macedo Sanchez, O., J. Surmont, C. Kissel, and C. Laj, New temporal constraints on the rotation of the Peruvian Central Andes obtained from paleomagnetism, *Geophys. Res. Lett., 19*, 1875-1878, 1992b.

MacFadden, B., F. Anaya, H. Perez, C. Naeser, P. Zeitler, and K. Campbell, Late Cenozoic paleomagnetism and chronology of Andean basins of Bolivia: Evidence for possible oroclinal bending, *J. Geol., 98*, 541-555, 1990.

MacFadden, B., F. Anaya and J.Argollo, Magnetic polarity stratigraphy of Inchasi: A Pliocene mammal-bearing locality from the Bolivian Andes deposited just before the Great American Interchange, *Earth Planet. Sci. Lett., 114*, 229-241, 1993.

MacFadden, B., F. Anaya, and C. Swisher III, Neogene paleomagnetism and oroclinal bending of the central Andes of Bolivia, *J. Geophys. Res., 100*, 8153-8167, 1995.

Mankinen, E., Revised paleomagnetic pole for the Sonoma volcanics, California, *Geophys. Res. Lett., 16*, 1081-1084, 1989.

Marret, R., The Late Cenozoic tectonic evolution of the Puna Plateau and adjacent foreland, northwestern Argentine Andes, Ph.D. thesis, Cornell Univ., Ithaca, N.Y., 1990.

Marret, R., R. Allmendinger, R. Alonso, and R. Drake, Late Cenozoic tectonic evolution of the Puna Plateau and adjacent foreland, northwestern Argentine Andes, *J. South Am. Earth Sci., 7*, 179-207, 1994.

May, S., and R. Butler, Paleomagnetism of the Puente Piedra Formation, central Perú, *Earth Planet. Sci. Lett., 72*, 205-218, 1985.

McFadden, P., and M. McElhinny, The combined analysis of remagnetization circles and direct observations in paleomagnetism, *Earth Planet. Sci. Lett., 87*, 161-172, 1988.

McFadden, P., R. Merrill, M. McElhinny, and S. Lee, Reversals of the Earth's magnetic field and temporal variations of the dynamo families, *J. Geophys. Res., 96*, 3923-3933, 1991.

McKenzie D., and J. Jackson, The relationship between strain rates, crustal thickening, palaeomagnetism, finite strain and fault movements within a deforming zone, *Earth Planet. Sci. Lett., 65*, 182-202, 1983.

Mégard, F., Cordilleran Andes and Marginal Andes: A review of Andean geology north of the Arica elbow (18°S), in *Circum-Pacific Belts and Evolution of the Pacific Ocean Basin, Geodyn. Ser.*, vol.18, edited by J. Monger and J. Francheteau, pp. 71-95, AGU, Washington, D. C., 1987.

Mon, R., and F. Hongn, Estructura del Ordovícico de la Puna, *Rev. Asoc. Geol. Argent., 42*, 31-38, 1988.

Mpodozis, C., and V. Ramos, The Andes of Chile and Argentina, in *Geology of the Andes and Its Relation to Hydrocarbon and Mineral Resources, Earth Sci. Ser.*, vol.11, edited by G. Eriksen et al., Circum-Pac. Counc. for Energy and Miner. Resour., pp. 59-90, Houston, Tex., 1990.

Ort, M., Eruptive processes and caldera formation in a nested downsag-collapse caldera: Cerro Panizos, Central Andes mountains, *J. Volcanol. Geotherm. Res., 56*, 221-252, 1993.

Oviedo, E., MAG88: Un sistema de computación para análisis de datos paleomagnéticos, Tesis Doctoral, Fac. de Cienc. Exactas y Nat., Univ. de Buenos Aires, Buenos Aires, 1989.

Pardo Casas, F., and P. Molnar, Relative motion of the Nazca (Farallon) and South America plates since Late Cretaceous time, *Tectonics* ,6, 233-248, 1987.

Park, R., *Geological Structures and Moving Plates*, 337 pp., Blackie, Glasgow, 1988.

Ré, G., and S. Barredo, Estudio magnetoestratigráfico y tasa de sedimentación del Grupo Iglesia en sus afloramientos aledaños a la localidad de Angualasto (Prov. de San Juan), in *Actas XII Congreso Geológico Argentino*, vol. 2, pp. 148-155, Mendoza, Argentina, 1993.

Richardson, A., and N. Watkins, Palaeomagnetism of Atlantic Islands: Fernando Noronha, *Nature, 215*, 1470-1473, 1967.

Roperch, P., and G. Carlier, Paleomagnetism of Mesozoic rocks from the central Andes of southern Peru: Importance of rotations in the development of the Bolivian Orocline, *J. Geophys. Res., 97*, 17,233-17,249, 1992.

Roperch, P., M. Fornari, and G. Hérail, A paleomagnetic study of the Altiplano (extended abstract), in *Symposium International Géodynamique Andine, 2nd.*, edited by ORSTOM, pp. 241-244, Oxford, England, 1993.

Rosembaum, J., Paleomagnetic directional dispertion produced by plastic deformation in a thick miocene welded tuff, southern Nevada: Implications for welding temperatures, *J. Geophys. Res., 91*, 12,817-12,834, 1986.

Scanlan, P., and P. Turner, Structural constraints on palaeomagnetic rotations south of the Arica bend, northern Chile: Implications for the Bolivian Orocline, *Tectonophysics, 205*, 141-154, 1992.

Scheuber, E., and K. Reutter, Magmatic arc tectonics in the Central Andes between 21° and 25°S, *Tectonophysics, 205*, 127-140, 1992.

Sempere, T., G. Hérail, J. Oller, and M.Bonhomme, Late Oligocene-early Miocene major tectonic crisis and related basins in Bolivia, *Geology, 18*, 946-949, 1990.

Singer, S., N. Puente, A. Sinito, and R. Allmendinger, Estudio paleomagnético de secuencias aflorantes al Este del Salar de Antofalla, Puna Austral, Argentina, in *Actas 7° Congreso Geológico de Chile*, vol. 2, pp. 1451-1455, Concepción, Chile, 1994.

Somoza, R., South American reference pole for the mid-Cretaceous: Further constraints in the interpretation of Andean paleomagnetic data, *Geology, 22*, 933-936, 1994.

Somoza, R., S. Singer, B. Coira, J. Vilas, J. Díaz, and P. Caffe, Ignimbritas del Complejo Volcánico Vilama (22.6°S-67°O), correlaciones paleomagnéticas y petrológicas, in *Actas 7° Congreso Geológico de Chile*, vol. 1, pp. 179-183, Concepción, Chile, 1994.

Tsunakawa, H., H. Tanaka, K. Amano, and M. Kono, Paleomagnetic study of Late Miocene and Early Pleistocene rocks from

Southern Peru, Central Andes, *J. Geomagn. Geoelectr., 39*, 477-486, 1987.

Turner, J., Descripción geológica de la hoja 3 ab, Mina Pirquitas, *Serv. Geol. Nac., Bol. 187*, 54 pp., Buenos Aires, Argentina, 1982.

Turner, J., and V. Mendez, Puna, in *Segundo Simposio de Geología Regional Argentina,* vol. 1, pp. 13-56, Acad. Nac. de Cienc., Córdoba, Argentina, 1979.

Vilas, J., Equipo para la desmagnetización de rocas para el estudio paleomagnético de las mismas, Tesis de Licenciatura, Fac. de Cienc. Exactas y Nat., Univ. de Buenos Aires, Buenos Aires, 1966.

Wolff, J., and J. Wright, Rheomorphism of welded tuffs. *J. volcanol. Geotherm. Res., 10*, 13-34, 1981.

B. Coira, Instituto de Geología y Minería,Universidad Nacional de Jujuy , (4600) S.S. de Jujuy, Argentina.

S. Singer and R. Somoza, Laboratorio de Paleomagnetismo Daniel Valencio, Departamento de Geología, Facultad de Ciencias Exactas y Naturales, Pabellón II, Ciudad Universitaria, (1428) Buenos Aires, Argentina. (e-mail: somoza@tango.gl.fcen.uba.ar)

(Received March 31, 1995; revised October 4, 1995;
accepted November 2, 1995.)